

## KING ISLAND CURRIE, Australia

WMO: 948510

Lat: 39.933S

Lon: 143.850E

Elev: 24

StdP: 101.04

Time Zone: 10.00 (AUS)

Period: 82-95

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 6.3        | 7.2 | 1.7                           | 4.3 | 9.0  | 2.6 | 4.6 | 9.3  | 19.8                  | 12.2 | 18.0 | 11.7 | 2.9                   | 50   | 0.681 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 6.0                    | 26.2            | 19.4 | 23.2 | 18.2 | 21.5 | 17.5 | 19.9                | 24.3 | 18.9 | 22.1 | 18.1 | 20.6 | 6.4                  | 20   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 18.4                            | 13.3 | 21.2 | 17.6 | 12.7 | 20.3 | 16.9 | 12.1 | 19.7 | 57.0          | 24.5 | 53.8 | 22.1 | 51.1 | 20.6 | 25.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 17.6          | 15.0 | 13.3 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |
| (5)               |      |      | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 13.7   | 16.4  | 17.0  | 16.4  | 14.9  | 13.1  | 11.2  | 10.7  | 11.0  | 11.4  | 12.6  | 14.2  | 15.4 |
|                                                                                  | DBStd       | 2.95   | 2.07  | 2.48  | 2.32  | 2.22  | 1.59  | 1.50  | 1.44  | 1.48  | 1.50  | 1.87  | 2.13  | 2.28 |
|                                                                                  | HDD10.0     | 27     | 0     | 0     | 0     | 0     | 1     | 6     | 10    | 6     | 3     | 1     | 0     | 0    |
|                                                                                  | HDD18.3     | 1752   | 68    | 52    | 71    | 105   | 164   | 216   | 238   | 227   | 208   | 177   | 128   | 98   |
|                                                                                  | CDD10.0     | 1363   | 198   | 195   | 198   | 148   | 95    | 41    | 31    | 38    | 45    | 83    | 125   | 167  |
|                                                                                  | CDD18.3     | 46     | 9     | 14    | 11    | 3     | 0     | 0     | 0     | 0     | 0     | 1     | 3     | 7    |
|                                                                                  | CDH23.3     | 211    | 43    | 74    | 31    | 6     | 0     | 0     | 0     | 0     | 1     | 2     | 14    | 41   |
|                                                                                  | CDH26.7     | 59     | 14    | 22    | 6     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 14   |
| Wind                                                                             | WSAvg       | 6.5    | 6.7   | 6.5   | 6.1   | 5.7   | 5.4   | 6.4   | 6.4   | 7.3   | 6.4   | 7.0   | 6.8   | 6.7  |
| Precipitation                                                                    | PrecAvg     | 888    | 34    | 35    | 56    | 67    | 99    | 99    | 130   | 114   | 84    | 73    | 62    | 46   |
|                                                                                  | PrecMax     | 1286   | 96    | 105   | 162   | 168   | 220   | 166   | 242   | 193   | 182   | 161   | 149   | 130  |
|                                                                                  | PrecMin     | 556    | 8     | 6     | 10    | 18    | 30    | 32    | 37    | 26    | 26    | 21    | 14    | 8    |
|                                                                                  | PrecStd     | 170    | 18    | 26    | 34    | 38    | 46    | 33    | 51    | 42    | 35    | 38    | 34    | 33   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 29.1  | 30.0  | 27.2  | 24.0  | 18.8  | 15.8  | 15.2  | 16.4  | 19.3  | 21.7  | 26.5  | 28.5 |
|                                                                                  |             | MCWB   | 21.1  | 19.9  | 18.6  | 17.8  | 15.3  | 13.2  | 12.6  | 13.0  | 14.7  | 15.7  | 18.8  | 19.7 |
|                                                                                  | 2%          | DB     | 24.1  | 25.9  | 23.2  | 21.1  | 17.2  | 14.7  | 14.5  | 15.1  | 16.5  | 18.7  | 22.0  | 22.7 |
|                                                                                  |             | MCWB   | 19.6  | 19.7  | 18.1  | 16.7  | 14.7  | 12.8  | 12.1  | 12.3  | 13.2  | 14.6  | 16.7  | 18.2 |
|                                                                                  | 5%          | DB     | 21.3  | 22.4  | 21.3  | 19.4  | 16.2  | 14.1  | 13.6  | 14.2  | 15.0  | 16.7  | 19.3  | 20.3 |
|                                                                                  |             | MCWB   | 17.8  | 18.0  | 17.6  | 16.2  | 14.1  | 12.4  | 11.5  | 11.8  | 12.6  | 13.8  | 15.9  | 17.2 |
|                                                                                  | 10%         | DB     | 19.7  | 20.5  | 19.7  | 18.0  | 15.5  | 13.6  | 13.0  | 13.4  | 14.0  | 15.5  | 17.5  | 18.7 |
|                                                                                  |             | MCWB   | 16.8  | 17.2  | 17.0  | 15.7  | 13.7  | 11.9  | 11.2  | 11.4  | 11.8  | 12.9  | 14.8  | 15.8 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 22.5  | 21.6  | 20.1  | 18.9  | 16.1  | 14.1  | 13.5  | 13.5  | 15.0  | 16.5  | 19.1  | 20.1 |
|                                                                                  |             | MCDB   | 27.6  | 27.4  | 24.1  | 22.3  | 17.8  | 14.9  | 14.7  | 15.5  | 18.6  | 20.2  | 24.0  | 25.6 |
|                                                                                  | 2%          | WB     | 19.5  | 19.9  | 18.9  | 17.5  | 15.2  | 13.4  | 12.5  | 12.8  | 13.6  | 15.0  | 17.4  | 18.7 |
|                                                                                  |             | MCDB   | 23.0  | 24.7  | 21.7  | 19.9  | 16.6  | 14.4  | 13.9  | 14.4  | 15.8  | 17.5  | 21.2  | 22.3 |
|                                                                                  | 5%          | WB     | 18.3  | 18.7  | 18.0  | 16.6  | 14.6  | 12.8  | 12.0  | 12.3  | 12.8  | 14.1  | 16.0  | 17.4 |
|                                                                                  |             | MCDB   | 20.8  | 21.6  | 20.4  | 18.6  | 15.9  | 13.9  | 13.2  | 13.8  | 14.6  | 16.2  | 18.8  | 20.0 |
|                                                                                  | 10%         | WB     | 17.2  | 17.7  | 17.3  | 15.8  | 14.0  | 12.1  | 11.5  | 11.7  | 12.1  | 13.4  | 15.1  | 16.3 |
|                                                                                  |             | MCDB   | 19.3  | 19.9  | 19.6  | 17.7  | 15.3  | 13.3  | 12.8  | 13.1  | 13.7  | 15.3  | 17.3  | 18.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.5   | 6.0   | 5.1   | 4.6   | 3.5   | 3.2   | 3.4   | 3.5   | 4.5   | 4.8   | 5.4   | 5.3  |
|                                                                                  |             | MCDBR  | 9.5   | 9.9   | 8.1   | 6.8   | 4.8   | 3.9   | 3.8   | 4.3   | 6.7   | 7.5   | 8.8   | 8.6  |
|                                                                                  |             | MCWBR  | 5.7   | 4.3   | 4.1   | 3.5   | 3.1   | 3.0   | 3.1   | 3.0   | 4.7   | 4.5   | 5.1   | 4.8  |
|                                                                                  | 5% WB       | MCDBR  | 8.5   | 8.4   | 7.0   | 6.0   | 3.9   | 3.4   | 3.3   | 3.9   | 6.2   | 6.7   | 7.8   | 8.1  |
|                                                                                  |             | MCWBR  | 5.5   | 4.3   | 4.1   | 3.7   | 3.2   | 3.1   | 3.1   | 3.0   | 4.7   | 4.7   | 4.8   | 4.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.354  | 0.347 | 0.336 | 0.328 | 0.317 | 0.316 | 0.314 | 0.326 | 0.346 | 0.342 | 0.345 | 0.343 |      |
|                                                                                  | taud        | 2.516  | 2.553 | 2.584 | 2.589 | 2.596 | 2.579 | 2.589 | 2.556 | 2.490 | 2.514 | 2.526 | 2.545 |      |
|                                                                                  | Ebn at Noon | 975    | 957   | 926   | 868   | 813   | 777   | 807   | 856   | 900   | 952   | 979   | 992   |      |
|                                                                                  | Edh at Noon | 110    | 101   | 90    | 78    | 67    | 63    | 66    | 79    | 97    | 104   | 108   | 108   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 6.82   | 5.88  | 4.47  | 3.08  | 2.02  | 1.58  | 1.74  | 2.59  | 3.80  | 5.25  | 6.19  | 6.89  |      |
|                                                                                  | RadStd      | 0.45   | 0.28  | 0.21  | 0.15  | 0.10  | 0.10  | 0.10  | 0.17  | 0.21  | 0.29  | 0.35  | 0.40  |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (8 neighbors) | +0.39 | N/A     | N/A    | +0.72   | +0.46 | +0.51 | N/A         | -118    | +129    | +16     |

Nomenclature: See separate page